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Dataset: Netflix Movies and TV Shows

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OBJECTIVE

To explore the Netflix dataset, perform data cleaning to handle missing and invalid values, create new features for analysis, and uncover insights through exploratory data analysis and visualization. The aim is to understand content trends, audience targeting through ratings, genre popularity, geographic production patterns, and the platform's growth over time.

DATASET OVERVIEW

The Netflix dataset contains information about TV shows and movies available on Netflix. Each row represents a single title and includes details such as: show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, description.

Key Details:

Time Period: Titles released from the 1940s up to 2021, with addition dates from 2008 to 2021

Data Size: The original dataset had 8,807 rows and 12 columns

Content Types: The dataset includes both 6,131 Movies and 2,676 TV Shows

Main Focus: Content metadata including country of production, release year, content rating, duration, and genres

LIBRARIES USED: pandas, matplotlib.pyplot, seaborn

DATA CLEANING PROCESS

The dataset was first loaded into Python using pandas. The file loaded properly. After loading the data, I checked the structure of the dataset, viewed the first few rows, checked the number of rows and columns, reviewed the data types, and identified missing values.

CLEANING SUMMARY

Issue Found Action Taken

Missing director (2,634 rows) was filled with "Unknown" because the data was too large to drop

Missing cast (825 rows) was filled with "Unknown" because the data was too large to drop

Missing country (831 rows) was filled with "Unknown" and was excluded from country-specific analysis later

Missing date_added (10 rows) was removed because the data was too small to impact analysis

Missing rating (4 rows) was removed because it was too small to impact analysis

Missing duration (3 rows) was removed because it was too small to impact analysis

Duplicates none was found

Invalid rating values ("74 min", "84 min", "66 min") was resolved automatically because these rows were among those dropped for missing date_added/duration

date_added was stored as text Converted to datetime

duration stored as mixed text ("90 min" / "2 Seasons") was splited into two numeric columns: duration_minutes (Movies) and duration_seasons (TV Shows)

Column names was already in clean lowercase snake_case and was confirmed, no changes needed

possible invalid years and duration outliers filtered released_year 1900-2025 and was flagged for duration outliers

FINAL CLEAN DATASET: 8794 rows, 16 columns (4 new columns 'duration_num, duration_type, duration_minutes, duration_seasons' were added)

DATA VISUALIZATION SUMMARY/KEY FINDINGS & KEY INSIGHTS

Seven virtualization charts were created for better understanding.

Chart 1: Top 10 Content-Producing Countries

Key Finding: The United States is the largest producer of Netflix content, followed by India and the United Kingdom.

Key Insight: This shows Netflix's strategy focuses on English-speaking markets and India, which have the biggest subscriber bases.

Chart 2: Distribution of Movies vs TV Shows

Key Finding: Movies account for approximately 70% of the catalog, while TV Shows make up 30%.

Key Insight: Netflix prioritizes films to quickly grow its library, but still maintains a strong collection of series to keep users engaged long-term.

Chart 3: Content Added to Netflix by Year

Key Finding: The number of titles added peaked between 2017 and 2019, followed by a sharp decline in 2020.

Key Insight: The peak reflects Netflix's rapid expansion. The 2020 drop was likely caused by COVID-19 production delays.

Chart 4: Distribution of Release Years

Key Finding: Most of the content on Netflix was released after 2010. Very little content is from before 1990.

Key Insight: Netflix focuses on modern and recent releases to appeal to current audience preferences.

Chart 5: Distribution of Content Ratings

Key Finding: TV-MA and TV-14 are the 2 most common ratings. Kids ratings like TV-Y and G are very rare.

Key Insight: Netflix's catalog is heavily targeted toward Teen and Adult viewers, with limited family/kids content.

Chart 6: Distribution of TV Show Seasons

Key Finding: The median TV show has only 1 season. Most shows have 1-2 seasons, with a few outliers having 10+ seasons.

Key Insight: Netflix prefers limited series and short-run shows. Long-running shows are uncommon on the platform.

Chart 7: Top 10 Most Common Genres

Key Finding: The top genres are International Movies, Dramas, and Comedies. Documentaries and Action & Adventure also rank high.

Key Insight: Netflix focuses on Drama and International content to appeal globally, with Comedy as a key driver for engagement.

RECOMMENDATIONS

1. Content Production & Geography: Increase investment in underrepresented regions like Africa, Middle East, and Latin America.
2. Content Type Balance: Increase original TV Shows and limited series.
3. Audience Targeting: Develop more Kids and Family-friendly content with ratings TV-Y, G, TV-Y7.
4. Genre Strategy: Continue investing in International Movies, Dramas, and Comedies but also expand Documentaries and Reality TV.
5. Release Strategy: Focus on acquiring and producing recent content from the last 5 years.
6. TV Show Format: Produce more 1-3 season limited series instead of trying to make 10+ season shows.
7. Post-COVID Recovery: Bring back 2019-level content production rate.